



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.C.A. (CBCS) Degree Examinations

August 2023

22MCA202 - ENTERPRISE JAVA

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **each Unit**.*

		Marks	BT*	CO*	PO*
Unit – I					
1. a)	What are strings in Java? Demonstrate immutability of String class and mutability of StringBuffer class with example.	10	L2	1	1
b)	Discuss packages in java. Explain the creation of a package and importing package with code example.	10	L2	1	1
2. a)	Distinguish between method overloading and method overriding with suitable example. Can final methods be overridden in subclasses?	10	L2	1	1
b)	How is multiple inheritance achieved in Java? Demonstrate interfaces in Java with suitable example.	10	L2	1	1
Unit – II					
3. a)	Explain exception handling mechanism in Java with example. Write a Java program to define multiple catch blocks.	10	L2	2	1
b)	How do you create user defined exceptions in java? Explain with example. Also differentiate between throw and throws in Java.	10	L2	2	1
4. a)	Describe the two methods of creating threads in Java with code example.	10	L2	2	1
b)	Discuss synchronization. Explain the mechanisms for achieving inter thread communication.	10	L3	2	2
Unit – III					
5. a)	Discuss about Object Input, Object Output streams and serialization in Java.	10	L2	3	1
b)	Explain reading and writing files in Java with example and also transient and volatile modifiers.	10	L2	3	1
6. a)	Explain the applet architecture with applet skeleton and an example program.	10	L2	3	1
b)	Discuss event driven programming in Java including delegation event model, event classes and event listener interfaces.	10	L2	3	1
Unit – IV					
7. a)	Explain about URL class and discuss the concept of TCP/ IP client sockets and TCP/IP server sockets.	10	L2	4	1
b)	Describe Remote Method Invocation in Java with server side implementation and client side usage.	10	L2	4	1
8. a)	Explain JDBC process steps for querying database with example and also processing result sets.	10	L2	4	1
b)	Give an overview of Java Beans with an example for developing a simple bean.	10	L2	4	1
Unit – V					
9. a)	Describe the basic servlet structure and its life cycle methods.	10	L2	5	1
b)	Discuss the various types of JSP tags.	10	L2	5	1
10. a)	How does a message driven bean work? Explain the life cycle of message driven bean.	10	L2	5	1
b)	Explain the concept of Spring Boot and the process of creating a Spring Boot project using Spring Initializer.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
